| Course Code | Course Name | Theory | Practical | Tutorial | Theory | Practical/ Oral | Tutorial | Total |
|---|---|---|---|---|---|---|---|---|
| ITC701 | AI and DS –II | 03 | -- | -- | 03 | -- | -- | 03 |

| Course Code | Course Name | Examination Scheme | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Theory Marks | | | | Term Work | Practical | Oral | Total |
| | | Internal assessment | | | End Sem. Exam | | | | |
| | | Test1 | Test 2 | Avg. of 2 Tests | | | | | |
| ITC701 | AI and DS – II | 20 | 20 | 20 | 80 | -- | -- | -- | 100 |

**Course Objectives:**

| Sr. No. | Course Objectives |
|---|---|
| The course aims: | |
| 1 | To model a decision making for a new problem in an uncertain domain. |
| 2 | To demonstrate Cognitive skills of Artificial Intelligence. |
| 3 | To become familiar with the basics of Fuzzy Logic and Fuzzy Systems. |
| 4 | To become familiar with Deep Learning Concepts and Architectures. |
| 5 | To define and apply metrics to measure the performance of various learning algorithms. |
| 6 | To enable students to analyze data science methods for real world problems. |

**Course Outcomes:**

| Sr. No. | Course Outcomes | Cognitive levels of attainment as per Bloom's Taxonomy |
|---|---|---|
| On successful completion, of course, learner/student will be able to: | | |
| 1 | Design models for reasoning with uncertainty as well as the use of unreliable information. | L1,L2,L3 |
| 2 | Analyze the process of building a Cognitive application. | L1,L2,L3,L4 |
| 3 | Design fuzzy controller system. | L1,L2,L3 |
| 4 | Apply learning concepts to develop real life applications. | L1,L2,L3 |
| 5 | Evaluate performance of learning algorithms. | L1,L2,L3,L4,L5 |
| 6 | Analyze current trends in Data Science. | L1,L2,L3,L4 |

**Prerequisite:** AI and DS - 1 (ITC604) , Data Mining & Business Intelligence (ITC601)

**DETAILED SYLLABUS:**

| Sr. No. | Module | Detailed Content | Hours | CO Mapping |
|---|---|---|---|---|

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|---|---|---|---|---|
| 0 | Prerequisite | Intelligent Agents, Search Techniques, Knowledge and Reasoning, Data Science, Machine Learning. | **02** | -- |
| I | Uncertainty | Uncertainty in AI, Inference using full joint distributions, Bayes Theorem, the semantics of Bayesian Networks, Inference in Bayesian networks, Decision Theory, Markov Decision Processes.<br>**Self-learning Topics:** Hidden Markov Model (HMM), Gaussian Mixture Model (GMM). | **04** | CO1 |
| II | Cognitive Computing | Foundation of Cognitive Computing, Design Principles for Cognitive Systems, Natural Language Processing in Support of a Cognitive System, Representing Knowledge in Taxonomies and Ontologies, Applying Advanced Analytics to Cognitive Computing, The Process of Building a Cognitive Application.<br>**Self-learning Topics:** Cognitive Systems such as IBM's Watson. | **06** | CO2 |
| III | Fuzzy Logic & Its Applications | Introduction to Fuzzy Sets, Properties of Fuzzy Sets, Operations on Fuzzy Sets, Fuzzy Membership Functions, Fuzzy Relations with Operations and its Properties, Fuzzy Composition: Max-Min Composition, Max-Product Composition, Defuzzification Methods, Architecture of Mamdani Type Fuzzy Control System, Design of Fuzzy Controllers like Domestic Shower Controller, Washing Machine Controller, Water Purifier Controller, etc.<br>**Self-learning Topics:** Other Fuzzy Composition Operations, Fuzzy Inference System (FIS) & ANFIS. | **07** | CO3 |
| IV | Introduction to Deep Learning | Introduction to Deep Learning, ANN, Machine Learning Vs Deep Learning, Working of Deep Learning; Convolutional Neural Network: Introduction, Components of CNN Architecture, Properties of CNN, Architectures of CNN, Applications of CNN, Recurrent Neural Network: Introduction, Simple RNN, LSTM Implementation, Deep RNN, Autoencoder: Introduction, Features, Types, Applications of Deep Learning.<br>**Self-learning Topics:** Restricted Boltzmann Machine (RBM). | **08** | CO4 |
| V | Advanced ML Classification Techniques | Ensemble Classifiers: Introduction to Ensemble Methods, Bagging, Boosting, Random forests, Improving classification accuracy of Class-Imbalanced Data.<br>Metrics for Evaluating Classifier Performance, Holdout Method and Random Subsampling, Cross-Validation, Bootstrap, Model Selection Using Statistical Tests of Significance, Comparing Classifiers Based on Cost–Benefit and ROC Curves.<br>**Self-learning Topics:** Introduction to ML (Revision), | **06** | CO4<br>CO5 |

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|---|---|---|---|---|
| | | Introduction to Reinforcement Learning. | | |
| VI | Trends and applications in Data Science | Data Science: applications and case studies, Data science for text, image, video, audio. Data science for Multimodal applications.<br><br>**Self-learning Topics:** ImageNet Large Scale Visual Recognition Challenge (ILSVRC). | **06** | CO6 |

**Text Books:**


1. Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Third Edition, Pearson Education.
2. Judith S. Hurwitz, Marcia Kaufman, Adrian Bowles, "Cognitive Computing and Big Data Analytics", Wiley India, 2015.
3. S.N. Sivanandam, S.N. Deepa, "Principles of Soft Computing", Wiley Publication.
4. Dr. S Lovelyn Rose, Dr. L Ashok Kumar, Dr. D Karthika Renuka, "Deep Learning Using Python", Wiley India, 2020.
5. B. Uma Maheshwari, R. Sujatha, "Introduction to Data Science Practical Approach with R and Python", Wiley India, 2021.
6. François Chollet, "Deep Learning with Python", Manning Publications, 2018.
7. Han J, Kamber M, Pei J, "Data Mining Concepts and Techniques", Third Edition, Morgan Kaufmann.


**References:**


1. Deepak Khemani, "A First Course in Artificial Intelligence", McGraw Hill Publication.
2. Ethem Alpaydin , "Introduction to Machine Learning", PHI Learning Pvt. Ltd.
3. Jon Krohn, Grant Beyleveld, Aglae Bassens, "Deep Learning Illustrated: A Visual, Interactive Guide to Artificial Intelligence", Pearson Education.
4. Prateek Joshi, "Artificial Intelligence with Python", Packt Publishing.


**Online References:**

| Sr. No. | Website Links |
|---|---|
| 1 | https://d2l.ai/index.html |
| 2 | https://onlinecourses.nptel.ac.in/noc20_cs62/preview |
| 3 | https://onlinecourses.nptel.ac.in/noc22_cs35/preview |
| 4 | https://www.coursera.org/specializations/deep-learning |
| 5 | https://onlinecourses.nptel.ac.in/noc22_cs56/preview |

**Assessment:**

**Internal Assessment for 20 marks**:

Consisting of **Two Compulsory Class Tests**

Approximately 40% to 50% of syllabus content must be covered in First test and remaining 40% to 50% of syllabus contents must be covered in second test.

**End Semester Examination**: Some guidelines for setting the question papers are as:

- Weightage of each module in end semester examination is expected to be/will be proportional to number of respective lecture hours mentioned in the syllabus.
- Question paper will comprise of total **six questions, each carrying 20 marks**.
- **Q.1** will be **compulsory** and should **cover maximum contents of the syllabus**.
- **Remaining question will be mixed in nature** (for example if Q.2 has part (a) from module 3 then part (b) will be from any other module. (Randomly selected from all the modules.)

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